NSCLC_Modeling_3

(Number of altered genes in the pathway for the patient) / (Total number of genes in that KEGG pathway).

Setups

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(glmnet)
```

```
Loading required package: Matrix

Loaded glmnet 4.1-10
```

```
library(survminer)
```

```
Loading required package: ggplot2
```

```
Warning: package 'ggplot2' was built under R version 4.4.3

Loading required package: ggpubr
```

```r
library(LTRCforests)
```

```
Attaching package: 'LTRCforests'

The following object is masked from 'package:Matrix':

    print

The following object is masked from 'package:base':

    print
```

```r
library(ggplot2)
library(survival)
```

```
Attaching package: 'survival'

The following object is masked from 'package:survminer':

    myeloma
```

```r
nsclc <- read.csv("nsclc.csv") # read in the dataset

# Adjusting factor variables
# Adjusting factor variables
nsclc <- nsclc %>% mutate(
  stage_f = as.factor(stage_f),
  smoke_f = as.factor(smoke_f),
  prior_med_f = as.factor(prior_med_to_msk_f),
  gender = ifelse(GENDER == "Female", 1 ,0) # change Female to 1, and male to 0
) %>% select(-c(
  "GENDER", "STAGE_CDM_DERIVED",
  "SMOKING_PREDICTIONS_3_CLASSES", "PRIOR_MED_TO_MSK"
```

```r
))

# 1. Define clinical columns to isolate the gene matrix

clinical_cols <- c(

  "X", "PATIENT_ID", "age_at_diagnosis",  "event",

  "entry_time", "exit_time", "smoke_f", "stage_f", "prior_med_to_msk_f",

  "prior_med_f", "gender"

)

# Identify genes by subtracting clinical columns from all column names
gene_cols <- setdiff(names(nsclc), clinical_cols)

# Create the matrix used in the loop
gene_matrix <- nsclc[, gene_cols]
```

Data Processing

```r
# install.packages("msigdbr")
library(msigdbr)
```

```
Warning: package 'msigdbr' was built under R version 4.4.3
```

```r
library(tibble)

# 1. Pull KEGG pathways from MSigDB
# C2 category, CP:KEGG subcategory contains all KEGG pathways
kegg_data <- msigdbr(
  species = "Homo sapiens",
  collection = "C2",
  subcollection = "CP:KEGG_LEGACY"
)
# 2. Convert to a list where each element is a pathway containing its gene symbols
kegg_list <- split(x = kegg_data$gene_symbol, f = kegg_data$gs_name)
```

```r
# 3. Create an empty list to store the fractional scores for each pathway
pathway_scores <- list()

# 4. Loop through each KEGG pathway to calculate the fraction
for (pathway_name in names(kegg_list)) {

  # Get the unique genes in this specific KEGG pathway
  pathway_genes <- unique(kegg_list[[pathway_name]])
  total_genes_in_pathway <- length(pathway_genes)

  # Find which of these pathway genes are actually measured in our NSCLC dataset
  # (gene_cols was defined in your Step 1)
  intersecting_genes <- intersect(pathway_genes, gene_cols)

  # If we have at least one gene from this pathway in our dataset:
  if (length(intersecting_genes) > 0) {

    # Sum the mutations (0s and 1s) across the intersecting genes for each patient
    # rowSums handles this perfectly across the isolated columns
    patient_mutations <- rowSums(gene_matrix[, intersecting_genes, drop = FALSE], na.rm = TRU

    # Calculate the fraction: (Altered genes) / (Total genes in KEGG pathway)
    fraction <- patient_mutations / total_genes_in_pathway

    # Store in our list
    pathway_scores[[pathway_name]] <- fraction
  }
}

# 5. Combine the list into a dataframe
pathway_matrix <- bind_cols(pathway_scores)

# 6. Filter out pathways with zero variance (no patients had mutations in them)
pathway_df <- colSums(pathway_matrix, na.rm = TRUE)
active_pathways <- names(pathway_df[pathway_df > 0])
pathway_matrix_filtered <- pathway_matrix[, active_pathways]

# 7. Recombine with your clinical data
nsclc_pathway <- bind_cols(nsclc[, clinical_cols], pathway_matrix_filtered)

cat("Total KEGG Pathways evaluated:", length(kegg_list), "\n")
```

```
Total KEGG Pathways evaluated: 186
```

```r
cat("Pathways with at least 1 mutation in cohort:", length(active_pathways), "\n")
```

```
Pathways with at least 1 mutation in cohort: 112
```

2. Prepare data for glmnet

```r
library(tidyr)
```

```
Attaching package: 'tidyr'

The following objects are masked from 'package:Matrix':

    expand, pack, unpack
```

```r
# 1. Create a complete dataset (dropping NAs to align x and y)
nsclc_path_complete <- nsclc_pathway %>% drop_na()

# 2. Isolate the predictor variables (removing survival/ID columns)
x_vars_path <- nsclc_path_complete %>%
  select(-X, -PATIENT_ID, -entry_time, -exit_time, -event)

# 3. Create the predictor matrix (handling dummy variables for clinical factors)
x_matrix_path <- model.matrix(~ . - 1, data = x_vars_path)

# 4. Create the Survival response object
y_surv_path <- Surv(
  time = nsclc_path_complete$entry_time,
  time2 = nsclc_path_complete$exit_time,
  event = nsclc_path_complete$event
)
```

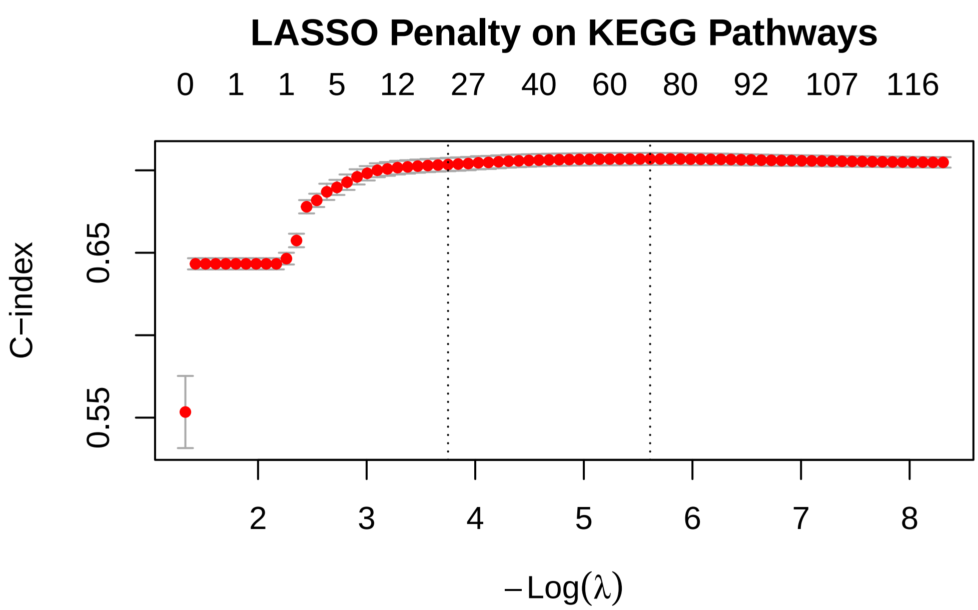
1. Cross-validation with Lasso

```r
set.seed(629)

# Fit the Cross-Validated Lasso Cox Model using Pathway features
cv_lasso_path <- cv.glmnet(
  x = x_matrix_path,
  y = y_surv_path,
  family = "cox",
  alpha = 1,
  type.measure = "C"
)
```

```
Warning: cox.fit: algorithm did not converge
Warning: cox.fit: algorithm did not converge
Warning: cox.fit: algorithm did not converge
Warning: cox.fit: algorithm did not converge
Warning: cox.fit: algorithm did not converge
Warning: cox.fit: algorithm did not converge
Warning: cox.fit: algorithm did not converge
Warning: cox.fit: algorithm did not converge
```

```r
plot(cv_lasso_path)
title("LASSO Penalty on KEGG Pathways", line = 2.5)
```



```r
cat("Optimal Lambda:", cv_lasso_path$lambda.min, "\n")
```

```
Optimal Lambda: 0.003659846
```

```r
cat("Max CV C-index (Pathways):", max(cv_lasso_path$cvm, na.rm = TRUE), "\n")
```

```
Max CV C-index (Pathways): 0.7069086
```

```r
# Extract the winning pathway features
lasso_coefs_path <- coef(cv_lasso_path, s = "lambda.min")
active_features_path <- rownames(lasso_coefs_path)[which(lasso_coefs_path != 0)]

cat("Features selected by LASSO:", length(active_features_path), "\n\n")
```

```
Features selected by LASSO: 70
```

```r
# View the selected features and weights
selected_weights_path <- lasso_coefs_path[active_features_path, , drop = FALSE]
print(selected_weights_path)
```

```
70 x 1 sparse Matrix of class "dgCMatrix"
                                                                  1
age_at_diagnosis                                      -7.861029e-03
smoke_fNever                                          -2.073891e-01
smoke_fUnknown                                         4.105759e-01
stage_fStage 4                                         1.083725e+00
prior_med_to_msk_fPrior medications to MSK             3.604857e-01
prior_med_to_msk_fUnknown                              1.720229e-01
prior_med_fPrior medications to MSK                    3.784428e-05
prior_med_fUnknown                                     1.114374e-15
gender                                                -2.784545e-01
KEGG_ABC_TRANSPORTERS                                  1.019671e+01
KEGG_ACUTE_MYELOID_LEUKEMIA                           -3.445398e+00
KEGG_ADHERENS_JUNCTION                                 2.672343e+00
KEGG_ADIPOCYTOKINE_SIGNALING_PATHWAY                   8.626610e+00
KEGG_ALLOGRAFT_REJECTION                               2.207477e+00
KEGG_AMYOTROPHIC_LATERAL_SCLEROSIS_ALS                 1.155487e+01
KEGG_ARRHYTHMOGENIC_RIGHT_VENTRICULAR_CARDIOMYOPATHY_ARVC -7.300011e+00
KEGG_ASTHMA                                            2.924857e-01
```

KEGG_AUTOIMMUNE_THYROID_DISEASE	2.527066e+00
KEGG_AXON_GUIDANCE	-5.240351e+00
KEGG_BASAL_TRANSCRIPTION_FACTORS	3.087483e+01
KEGG_BASE_EXCISION_REPAIR	-9.323404e-01
KEGG_CHRONIC_MYELOID_LEUKEMIA	5.581839e-01
KEGG_CITRATE_CYCLE_TCA_CYCLE	4.983306e+00
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM	-5.325298e+00
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION	-6.406584e+00
KEGG_CYTOSOLIC_DNA_SENSING_PATHWAY	5.417635e+00
KEGG_DORSO_VENTRAL_AXIS_FORMATION	-3.065050e-01
KEGG_ENDOCYTOSIS	-5.915325e+00
KEGG_EPITHELIAL_CELL_SIGNALING_IN_HELICOBACTER_PYLORI_INFECTION	-1.025308e+01
KEGG_ERBB_SIGNALING_PATHWAY	-1.109566e+00
KEGG_FC_EPSILON_RI_SIGNALING_PATHWAY	8.423948e+00
KEGG_HEDGEHOG_SIGNALING_PATHWAY	-7.485981e+00
KEGG_HUNTINGTONS_DISEASE	1.077553e+01
KEGG_HYPERTROPHIC_CARDIOMYOPATHY_HCM	-1.439868e+01
KEGG_INSULIN_SIGNALING_PATHWAY	-1.187150e+01
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION	-4.086068e-01
KEGG_LYSINE_DEGRADATION	-2.169275e+00
KEGG_MAPK_SIGNALING_PATHWAY	1.570607e+00
KEGG_MTOR_SIGNALING_PATHWAY	5.595486e+00
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY	3.283607e+00
KEGG_NICOTINATE_AND_NICOTINAMIDE_METABOLISM	1.428094e+00
KEGG_NOD_LIKE_RECEPTOR_SIGNALING_PATHWAY	-3.314314e+00
KEGG_NON_HOMOLOGOUS_END_JOINING	-2.842538e+00
KEGG_NON_SMALL_CELL_LUNG_CANCER	6.245262e-04
KEGG_NOTCH_SIGNALING_PATHWAY	-2.422637e+00
KEGG_NUCLEOTIDE_EXCISION_REPAIR	-2.479176e+00
KEGG_OOCYTE_MEIOSIS	-8.007483e+00
KEGG_OXIDATIVE_PHOSPHORYLATION	-5.502208e+01
KEGG_P53_SIGNALING_PATHWAY	2.902123e+00
KEGG_PANCREATIC_CANCER	-2.129061e+00
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION	-7.176032e+00
KEGG_PPAR_SIGNALING_PATHWAY	5.020286e+00
KEGG_PRIMARY_IMMUNODEFICIENCY	-5.409335e+00
KEGG_PRION_DISEASES	4.546445e-01
KEGG_REGULATION_OF_AUTOPHAGY	-8.794889e+00
KEGG_RENAL_CELL_CARCINOMA	1.623141e+00
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHWAY	-1.549052e+01
KEGG_RNA_DEGRADATION	-1.608050e+01
KEGG_SMALL_CELL_LUNG_CANCER	2.114554e+00
KEGG_SPLICEOSOME	-1.462125e+01

```
KEGG_STEROID_HORMONE_BIOSYNTHESIS                              3.181355e+00
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS                              1.479156e+00
KEGG_T_CELL_RECEPTOR_SIGNALING_PATHWAY                         2.045540e+00
KEGG_TGF_BETA_SIGNALING_PATHWAY                                6.205066e+00
KEGG_TIGHT_JUNCTION                                            1.579978e+01
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY                     -5.002934e+00
KEGG_TYPE_II_DIABETES_MELLITUS                                -1.845250e+00
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS                            1.995743e+01
KEGG_VIBRIO_CHOLERAE_INFECTION                                 3.824205e+00
KEGG_VIRAL_MYOCARDITIS                                        -4.692830e-01
```

Post-lasso Unpenalized Cox Model

```r
# 1. Identify ONLY the pathways that LASSO selected
selected_pathways <- active_features_path[active_features_path %in% colnames(pathway_matrix_i

# --- DIAGNOSTIC CHECK ---
# Check if any selected pathways have essentially 0 variance in the COMPLETE dataset
# If a pathway has only 0s, scale() will fail and coxph will crash.
pathway_variances <- apply(nsclc_path_complete[, selected_pathways, drop=FALSE], 2, var)
valid_pathways <- names(pathway_variances[pathway_variances > 0])

if(length(valid_pathways) < length(selected_pathways)) {
  cat("Dropped", length(selected_pathways) - length(valid_pathways),
      "pathways due to zero variance in the clean dataset.\n")
}

# 2. Build final data frame
final_model_data_path <- nsclc_path_complete %>%
  select(
    entry_time, exit_time, event,
    age_at_diagnosis, smoke_f, stage_f, prior_med_f, gender,
    all_of(valid_pathways)
  ) %>%
  # FIX 1: Standardize the continuous pathway fractions (Z-scores)
  # This makes the optimization algorithm much happier
  mutate(across(all_of(valid_pathways), ~ as.numeric(scale(.))))

# 3. Run the final Cox model
clean_cox_model_path <- coxph(
  Surv(entry_time, exit_time, event) ~ .,
```

```r
  data = final_model_data_path,
  # FIX 2: Give the algorithm 100 iterations instead of the default 20
  control = coxph.control(iter.max = 100)
)

# Check the output (and check if the warning is gone!)
summary(clean_cox_model_path)
```

```
Call:
coxph(formula = Surv(entry_time, exit_time, event) ~ ., data = final_model_data_path,
    control = coxph.control(iter.max = 100))

  n= 7802, number of events= 3971

                                                                 coef
age_at_diagnosis                                            -0.009368
smoke_fNever                                                -0.219969
smoke_fUnknown                                               0.402805
stage_fStage 4                                               1.108423
prior_med_fPrior medications to MSK                          0.385568
prior_med_fUnknown                                           0.204707
gender                                                      -0.293705
KEGG_ABC_TRANSPORTERS                                        0.042880
KEGG_ACUTE_MYELOID_LEUKEMIA                                 -0.107014
KEGG_ADHERENS_JUNCTION                                       0.117195
KEGG_ADIPOCYTOKINE_SIGNALING_PATHWAY                         0.077345
KEGG_ALLOGRAFT_REJECTION                                     0.011415
KEGG_AMYOTROPHIC_LATERAL_SCLEROSIS_ALS                       0.120250
KEGG_ARRHYTHMOGENIC_RIGHT_VENTRICULAR_CARDIOMYOPATHY_ARVC   -0.028095
KEGG_ASTHMA                                                        NA
KEGG_AUTOIMMUNE_THYROID_DISEASE                              0.018354
KEGG_AXON_GUIDANCE                                          -0.065042
KEGG_BASAL_TRANSCRIPTION_FACTORS                             0.015376
KEGG_BASE_EXCISION_REPAIR                                   -0.011678
KEGG_CHRONIC_MYELOID_LEUKEMIA                                0.092116
KEGG_CITRATE_CYCLE_TCA_CYCLE                                 0.059331
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM                     -0.042925
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION                 -0.030452
KEGG_CYTOSOLIC_DNA_SENSING_PATHWAY                           0.043315
KEGG_DORSO_VENTRAL_AXIS_FORMATION                            0.018947
KEGG_ENDOCYTOSIS                                            -0.023622
KEGG_EPITHELIAL_CELL_SIGNALING_IN_HELICOBACTER_PYLORI_INFECTION -0.130378
```

KEGG_ERBB_SIGNALING_PATHWAY	-0.037681
KEGG_FC_EPSILON_RI_SIGNALING_PATHWAY	0.122948
KEGG_HEDGEHOG_SIGNALING_PATHWAY	-0.043153
KEGG_HUNTINGTONS_DISEASE	0.033019
KEGG_HYPERTROPHIC_CARDIOMYOPATHY_HCM	-0.020408
KEGG_INSULIN_SIGNALING_PATHWAY	-0.197390
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION	-0.012695
KEGG_LYSINE_DEGRADATION	-0.020982
KEGG_MAPK_SIGNALING_PATHWAY	0.049270
KEGG_MTOR_SIGNALING_PATHWAY	0.148877
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY	0.046309
KEGG_NICOTINATE_AND_NICOTINAMIDE_METABOLISM	0.009223
KEGG_NOD_LIKE_RECEPTOR_SIGNALING_PATHWAY	-0.020809
KEGG_NON_HOMOLOGOUS_END_JOINING	-0.034512
KEGG_NON_SMALL_CELL_LUNG_CANCER	0.102230
KEGG_NOTCH_SIGNALING_PATHWAY	-0.082618
KEGG_NUCLEOTIDE_EXCISION_REPAIR	-0.018137
KEGG_OOCYTE_MEIOSIS	-0.042810
KEGG_OXIDATIVE_PHOSPHORYLATION	-0.081358
KEGG_P53_SIGNALING_PATHWAY	0.042807
KEGG_PANCREATIC_CANCER	-0.232685
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION	-0.069797
KEGG_PPAR_SIGNALING_PATHWAY	0.009109
KEGG_PRIMARY_IMMUNODEFICIENCY	-0.064456
KEGG_PRION_DISEASES	0.021281
KEGG_REGULATION_OF_AUTOPHAGY	-0.037733
KEGG_RENAL_CELL_CARCINOMA	0.063002
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHWAY	-0.076287
KEGG_RNA_DEGRADATION	-0.043613
KEGG_SMALL_CELL_LUNG_CANCER	0.037318
KEGG_SPLICEOSOME	-0.030529
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.013109
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS	0.015286
KEGG_T_CELL_RECEPTOR_SIGNALING_PATHWAY	0.034983
KEGG_TGF_BETA_SIGNALING_PATHWAY	0.042475
KEGG_TIGHT_JUNCTION	0.087784
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY	-0.080325
KEGG_TYPE_II_DIABETES_MELLITUS	0.009034
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS	0.086477
KEGG_VIBRIO_CHOLERAE_INFECTION	0.019498
KEGG_VIRAL_MYOCARDITIS	0.009402
	exp(coef)
age_at_diagnosis	0.990676

smoke_fNever	0.802544
smoke_fUnknown	1.496014
stage_fStage 4	3.029578
prior_med_fPrior medications to MSK	1.470449
prior_med_fUnknown	1.227165
gender	0.745496
KEGG_ABC_TRANSPORTERS	1.043812
KEGG_ACUTE_MYELOID_LEUKEMIA	0.898513
KEGG_ADHERENS_JUNCTION	1.124338
KEGG_ADIPOCYTOKINE_SIGNALING_PATHWAY	1.080415
KEGG_ALLOGRAFT_REJECTION	1.011480
KEGG_AMYOTROPHIC_LATERAL_SCLEROSIS_ALS	1.127779
KEGG_ARRHYTHMOGENIC_RIGHT_VENTRICULAR_CARDIOMYOPATHY_ARVC	0.972296
KEGG_ASTHMA	NA
KEGG_AUTOIMMUNE_THYROID_DISEASE	1.018524
KEGG_AXON_GUIDANCE	0.937028
KEGG_BASAL_TRANSCRIPTION_FACTORS	1.015495
KEGG_BASE_EXCISION_REPAIR	0.988389
KEGG_CHRONIC_MYELOID_LEUKEMIA	1.096492
KEGG_CITRATE_CYCLE_TCA_CYCLE	1.061126
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM	0.957983
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION	0.970007
KEGG_CYTOSOLIC_DNA_SENSING_PATHWAY	1.044267
KEGG_DORSO_VENTRAL_AXIS_FORMATION	1.019128
KEGG_ENDOCYTOSIS	0.976654
KEGG_EPITHELIAL_CELL_SIGNALING_IN_HELICOBACTER_PYLORI_INFECTION	0.877763
KEGG_ERBB_SIGNALING_PATHWAY	0.963020
KEGG_FC_EPSILON_RI_SIGNALING_PATHWAY	1.130826
KEGG_HEDGEHOG_SIGNALING_PATHWAY	0.957764
KEGG_HUNTINGTONS_DISEASE	1.033570
KEGG_HYPERTROPHIC_CARDIOMYOPATHY_HCM	0.979799
KEGG_INSULIN_SIGNALING_PATHWAY	0.820870
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION	0.987386
KEGG_LYSINE_DEGRADATION	0.979237
KEGG_MAPK_SIGNALING_PATHWAY	1.050504
KEGG_MTOR_SIGNALING_PATHWAY	1.160531
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY	1.047398
KEGG_NICOTINATE_AND_NICOTINAMIDE_METABOLISM	1.009266
KEGG_NOD_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.979406
KEGG_NON_HOMOLOGOUS_END_JOINING	0.966077
KEGG_NON_SMALL_CELL_LUNG_CANCER	1.107638
KEGG_NOTCH_SIGNALING_PATHWAY	0.920703
KEGG_NUCLEOTIDE_EXCISION_REPAIR	0.982027

KEGG_OOCYTE_MEIOSIS	0.958094
KEGG_OXIDATIVE_PHOSPHORYLATION	0.921863
KEGG_P53_SIGNALING_PATHWAY	1.043736
KEGG_PANCREATIC_CANCER	0.792403
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION	0.932583
KEGG_PPAR_SIGNALING_PATHWAY	1.009151
KEGG_PRIMARY_IMMUNODEFICIENCY	0.937577
KEGG_PRION_DISEASES	1.021510
KEGG_REGULATION_OF_AUTOPHAGY	0.962970
KEGG_RENAL_CELL_CARCINOMA	1.065029
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.926550
KEGG_RNA_DEGRADATION	0.957325
KEGG_SMALL_CELL_LUNG_CANCER	1.038023
KEGG_SPLICEOSOME	0.969932
KEGG_STEROID_HORMONE_BIOSYNTHESIS	1.013195
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS	1.015404
KEGG_T_CELL_RECEPTOR_SIGNALING_PATHWAY	1.035602
KEGG_TGF_BETA_SIGNALING_PATHWAY	1.043390
KEGG_TIGHT_JUNCTION	1.091752
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.922816
KEGG_TYPE_II_DIABETES_MELLITUS	1.009075
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS	1.090326
KEGG_VIBRIO_CHOLERAE_INFECTION	1.019689
KEGG_VIRAL_MYOCARDITIS	1.009447
	se(coef)
age_at_diagnosis	0.001531
smoke_fNever	0.043603
smoke_fUnknown	0.067198
stage_fStage 4	0.034494
prior_med_fPrior medications to MSK	0.043543
prior_med_fUnknown	0.053455
gender	0.033074
KEGG_ABC_TRANSPORTERS	0.018041
KEGG_ACUTE_MYELOID_LEUKEMIA	0.044742
KEGG_ADHERENS_JUNCTION	0.034912
KEGG_ADIPOCYTOKINE_SIGNALING_PATHWAY	0.022310
KEGG_ALLOGRAFT_REJECTION	0.020414
KEGG_AMYOTROPHIC_LATERAL_SCLEROSIS_ALS	0.050747
KEGG_ARRHYTHMOGENIC_RIGHT_VENTRICULAR_CARDIOMYOPATHY_ARVC	0.023699
KEGG_ASTHMA	0.000000
KEGG_AUTOIMMUNE_THYROID_DISEASE	0.017106
KEGG_AXON_GUIDANCE	0.024745
KEGG_BASAL_TRANSCRIPTION_FACTORS	0.011590

KEGG_BASE_EXCISION_REPAIR	0.020974
KEGG_CHRONIC_MYELOID_LEUKEMIA	0.055598
KEGG_CITRATE_CYCLE_TCA_CYCLE	0.022448
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM	0.017289
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION	0.027632
KEGG_CYTOSOLIC_DNA_SENSING_PATHWAY	0.020621
KEGG_DORSO_VENTRAL_AXIS_FORMATION	0.036048
KEGG_ENDOCYTOSIS	0.027669
KEGG_EPITHELIAL_CELL_SIGNALING_IN_HELICOBACTER_PYLORI_INFECTION	0.031628
KEGG_ERBB_SIGNALING_PATHWAY	0.042439
KEGG_FC_EPSILON_RI_SIGNALING_PATHWAY	0.049700
KEGG_HEDGEHOG_SIGNALING_PATHWAY	0.017773
KEGG_HUNTINGTONS_DISEASE	0.047707
KEGG_HYPERTROPHIC_CARDIOMYOPATHY_HCM	0.017315
KEGG_INSULIN_SIGNALING_PATHWAY	0.050179
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION	0.019347
KEGG_LYSINE_DEGRADATION	0.016703
KEGG_MAPK_SIGNALING_PATHWAY	0.031110
KEGG_MTOR_SIGNALING_PATHWAY	0.031591
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY	0.049336
KEGG_NICOTINATE_AND_NICOTINAMIDE_METABOLISM	0.017258
KEGG_NOD_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.018745
KEGG_NON_HOMOLOGOUS_END_JOINING	0.016029
KEGG_NON_SMALL_CELL_LUNG_CANCER	0.062603
KEGG_NOTCH_SIGNALING_PATHWAY	0.031370
KEGG_NUCLEOTIDE_EXCISION_REPAIR	0.021003
KEGG_OOCYTE_MEIOSIS	0.018865
KEGG_OXIDATIVE_PHOSPHORYLATION	0.025705
KEGG_P53_SIGNALING_PATHWAY	0.027189
KEGG_PANCREATIC_CANCER	0.056691
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION	0.026782
KEGG_PPAR_SIGNALING_PATHWAY	0.016789
KEGG_PRIMARY_IMMUNODEFICIENCY	0.020334
KEGG_PRION_DISEASES	0.020577
KEGG_REGULATION_OF_AUTOPHAGY	0.015853
KEGG_RENAL_CELL_CARCINOMA	0.045338
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.021786
KEGG_RNA_DEGRADATION	0.016641
KEGG_SMALL_CELL_LUNG_CANCER	0.040085
KEGG_SPLICEOSOME	0.016806
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.019078
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS	0.026245
KEGG_T_CELL_RECEPTOR_SIGNALING_PATHWAY	0.040136

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KEGG_TGF_BETA_SIGNALING_PATHWAY                                 0.028316
KEGG_TIGHT_JUNCTION                                             0.030484
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY                       0.040741
KEGG_TYPE_II_DIABETES_MELLITUS                                  0.033458
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS                             0.017617
KEGG_VIBRIO_CHOLERAE_INFECTION                                  0.019362
KEGG_VIRAL_MYOCARDITIS                                          0.022139
                                                                     z Pr(>|z|)
age_at_diagnosis                                                -6.120 9.35e-10
smoke_fNever                                                    -5.045 4.54e-07
smoke_fUnknown                                                   5.994 2.04e-09
stage_fStage 4                                                  32.134  < 2e-16
prior_med_fPrior medications to MSK                              8.855  < 2e-16
prior_med_fUnknown                                               3.830 0.000128
gender                                                          -8.880  < 2e-16
KEGG_ABC_TRANSPORTERS                                            2.377 0.017462
KEGG_ACUTE_MYELOID_LEUKEMIA                                     -2.392 0.016765
KEGG_ADHERENS_JUNCTION                                           3.357 0.000788
KEGG_ADIPOCYTOKINE_SIGNALING_PATHWAY                             3.467 0.000527
KEGG_ALLOGRAFT_REJECTION                                         0.559 0.576050
KEGG_AMYOTROPHIC_LATERAL_SCLEROSIS_ALS                           2.370 0.017807
KEGG_ARRHYTHMOGENIC_RIGHT_VENTRICULAR_CARDIOMYOPATHY_ARVC       -1.185 0.235824
KEGG_ASTHMA                                                         NA       NA
KEGG_AUTOIMMUNE_THYROID_DISEASE                                  1.073 0.283271
KEGG_AXON_GUIDANCE                                              -2.629 0.008576
KEGG_BASAL_TRANSCRIPTION_FACTORS                                 1.327 0.184599
KEGG_BASE_EXCISION_REPAIR                                       -0.557 0.577657
KEGG_CHRONIC_MYELOID_LEUKEMIA                                    1.657 0.097554
KEGG_CITRATE_CYCLE_TCA_CYCLE                                     2.643 0.008217
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM                         -2.483 0.013035
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION                     -1.102 0.270449
KEGG_CYTOSOLIC_DNA_SENSING_PATHWAY                               2.101 0.035680
KEGG_DORSO_VENTRAL_AXIS_FORMATION                                0.526 0.599160
KEGG_ENDOCYTOSIS                                                -0.854 0.393236
KEGG_EPITHELIAL_CELL_SIGNALING_IN_HELICOBACTER_PYLORI_INFECTION -4.122 3.75e-05
KEGG_ERBB_SIGNALING_PATHWAY                                     -0.888 0.374603
KEGG_FC_EPSILON_RI_SIGNALING_PATHWAY                             2.474 0.013369
KEGG_HEDGEHOG_SIGNALING_PATHWAY                                 -2.428 0.015181
KEGG_HUNTINGTONS_DISEASE                                         0.692 0.488856
KEGG_HYPERTROPHIC_CARDIOMYOPATHY_HCM                            -1.179 0.238525
KEGG_INSULIN_SIGNALING_PATHWAY                                  -3.934 8.36e-05
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION               -0.656 0.511721
KEGG_LYSINE_DEGRADATION                                         -1.256 0.209045
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KEGG_MAPK_SIGNALING_PATHWAY	1.584	0.113259
KEGG_MTOR_SIGNALING_PATHWAY	4.713	2.44e-06
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY	0.939	0.347918
KEGG_NICOTINATE_AND_NICOTINAMIDE_METABOLISM	0.534	0.593049
KEGG_NOD_LIKE_RECEPTOR_SIGNALING_PATHWAY	-1.110	0.266960
KEGG_NON_HOMOLOGOUS_END_JOINING	-2.153	0.031317
KEGG_NON_SMALL_CELL_LUNG_CANCER	1.633	0.102469
KEGG_NOTCH_SIGNALING_PATHWAY	-2.634	0.008448
KEGG_NUCLEOTIDE_EXCISION_REPAIR	-0.864	0.387840
KEGG_OOCYTE_MEIOSIS	-2.269	0.023249
KEGG_OXIDATIVE_PHOSPHORYLATION	-3.165	0.001551
KEGG_P53_SIGNALING_PATHWAY	1.574	0.115394
KEGG_PANCREATIC_CANCER	-4.104	4.05e-05
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION	-2.606	0.009158
KEGG_PPAR_SIGNALING_PATHWAY	0.543	0.587413
KEGG_PRIMARY_IMMUNODEFICIENCY	-3.170	0.001525
KEGG_PRION_DISEASES	1.034	0.301019
KEGG_REGULATION_OF_AUTOPHAGY	-2.380	0.017302
KEGG_RENAL_CELL_CARCINOMA	1.390	0.164642
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHWAY	-3.502	0.000463
KEGG_RNA_DEGRADATION	-2.621	0.008773
KEGG_SMALL_CELL_LUNG_CANCER	0.931	0.351875
KEGG_SPLICEOSOME	-1.817	0.069284
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.687	0.492015
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS	0.582	0.560272
KEGG_T_CELL_RECEPTOR_SIGNALING_PATHWAY	0.872	0.383418
KEGG_TGF_BETA_SIGNALING_PATHWAY	1.500	0.133611
KEGG_TIGHT_JUNCTION	2.880	0.003981
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY	-1.972	0.048654
KEGG_TYPE_II_DIABETES_MELLITUS	0.270	0.787154
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS	4.909	9.16e-07
KEGG_VIBRIO_CHOLERAE_INFECTION	1.007	0.313941
KEGG_VIRAL_MYOCARDITIS	0.425	0.671064

age_at_diagnosis	***
smoke_fNever	***
smoke_fUnknown	***
stage_fStage 4	***
prior_med_fPrior medications to MSK	***
prior_med_fUnknown	***
gender	***
KEGG_ABC_TRANSPORTERS	*
KEGG_ACUTE_MYELOID_LEUKEMIA	*

KEGG_ADHERENS_JUNCTION	***
KEGG_ADIPOCYTOKINE_SIGNALING_PATHWAY	***
KEGG_ALLOGRAFT_REJECTION	
KEGG_AMYOTROPHIC_LATERAL_SCLEROSIS_ALS	*
KEGG_ARRHYTHMOGENIC_RIGHT_VENTRICULAR_CARDIOMYOPATHY_ARVC	
KEGG_ASTHMA	
KEGG_AUTOIMMUNE_THYROID_DISEASE	
KEGG_AXON_GUIDANCE	**
KEGG_BASAL_TRANSCRIPTION_FACTORS	
KEGG_BASE_EXCISION_REPAIR	
KEGG_CHRONIC_MYELOID_LEUKEMIA	.
KEGG_CITRATE_CYCLE_TCA_CYCLE	**
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM	*
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION	
KEGG_CYTOSOLIC_DNA_SENSING_PATHWAY	*
KEGG_DORSO_VENTRAL_AXIS_FORMATION	
KEGG_ENDOCYTOSIS	
KEGG_EPITHELIAL_CELL_SIGNALING_IN_HELICOBACTER_PYLORI_INFECTION	***
KEGG_ERBB_SIGNALING_PATHWAY	
KEGG_FC_EPSILON_RI_SIGNALING_PATHWAY	*
KEGG_HEDGEHOG_SIGNALING_PATHWAY	*
KEGG_HUNTINGTONS_DISEASE	
KEGG_HYPERTROPHIC_CARDIOMYOPATHY_HCM	
KEGG_INSULIN_SIGNALING_PATHWAY	***
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION	
KEGG_LYSINE_DEGRADATION	
KEGG_MAPK_SIGNALING_PATHWAY	
KEGG_MTOR_SIGNALING_PATHWAY	***
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY	
KEGG_NICOTINATE_AND_NICOTINAMIDE_METABOLISM	
KEGG_NOD_LIKE_RECEPTOR_SIGNALING_PATHWAY	
KEGG_NON_HOMOLOGOUS_END_JOINING	*
KEGG_NON_SMALL_CELL_LUNG_CANCER	
KEGG_NOTCH_SIGNALING_PATHWAY	**
KEGG_NUCLEOTIDE_EXCISION_REPAIR	
KEGG_OOCYTE_MEIOSIS	*
KEGG_OXIDATIVE_PHOSPHORYLATION	**
KEGG_P53_SIGNALING_PATHWAY	
KEGG_PANCREATIC_CANCER	***
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION	**
KEGG_PPAR_SIGNALING_PATHWAY	
KEGG_PRIMARY_IMMUNODEFICIENCY	**
KEGG_PRION_DISEASES	

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KEGG_REGULATION_OF_AUTOPHAGY                                 *
KEGG_RENAL_CELL_CARCINOMA
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHWAY                   ***
KEGG_RNA_DEGRADATION                                         **
KEGG_SMALL_CELL_LUNG_CANCER
KEGG_SPLICEOSOME                                             .
KEGG_STEROID_HORMONE_BIOSYNTHESIS
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS
KEGG_T_CELL_RECEPTOR_SIGNALING_PATHWAY
KEGG_TGF_BETA_SIGNALING_PATHWAY
KEGG_TIGHT_JUNCTION                                          **
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY                    *
KEGG_TYPE_II_DIABETES_MELLITUS
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS                          ***
KEGG_VIBRIO_CHOLERAE_INFECTION
KEGG_VIRAL_MYOCARDITIS
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                                             exp(coef)
age_at_diagnosis                                                0.9907
smoke_fNever                                                    0.8025
smoke_fUnknown                                                  1.4960
stage_fStage 4                                                  3.0296
prior_med_fPrior medications to MSK                             1.4704
prior_med_fUnknown                                              1.2272
gender                                                          0.7455
KEGG_ABC_TRANSPORTERS                                           1.0438
KEGG_ACUTE_MYELOID_LEUKEMIA                                     0.8985
KEGG_ADHERENS_JUNCTION                                          1.1243
KEGG_ADIPOCYTOKINE_SIGNALING_PATHWAY                            1.0804
KEGG_ALLOGRAFT_REJECTION                                        1.0115
KEGG_AMYOTROPHIC_LATERAL_SCLEROSIS_ALS                          1.1278
KEGG_ARRHYTHMOGENIC_RIGHT_VENTRICULAR_CARDIOMYOPATHY_ARVC       0.9723
KEGG_ASTHMA                                                         NA
KEGG_AUTOIMMUNE_THYROID_DISEASE                                 1.0185
KEGG_AXON_GUIDANCE                                              0.9370
KEGG_BASAL_TRANSCRIPTION_FACTORS                                1.0155
KEGG_BASE_EXCISION_REPAIR                                       0.9884
KEGG_CHRONIC_MYELOID_LEUKEMIA                                   1.0965
KEGG_CITRATE_CYCLE_TCA_CYCLE                                    1.0611
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM                         0.9580
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION                     0.9700
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KEGG_CYTOSOLIC_DNA_SENSING_PATHWAY	1.0443
KEGG_DORSO_VENTRAL_AXIS_FORMATION	1.0191
KEGG_ENDOCYTOSIS	0.9767
KEGG_EPITHELIAL_CELL_SIGNALING_IN_HELICOBACTER_PYLORI_INFECTION	0.8778
KEGG_ERBB_SIGNALING_PATHWAY	0.9630
KEGG_FC_EPSILON_RI_SIGNALING_PATHWAY	1.1308
KEGG_HEDGEHOG_SIGNALING_PATHWAY	0.9578
KEGG_HUNTINGTONS_DISEASE	1.0336
KEGG_HYPERTROPHIC_CARDIOMYOPATHY_HCM	0.9798
KEGG_INSULIN_SIGNALING_PATHWAY	0.8209
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION	0.9874
KEGG_LYSINE_DEGRADATION	0.9792
KEGG_MAPK_SIGNALING_PATHWAY	1.0505
KEGG_MTOR_SIGNALING_PATHWAY	1.1605
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY	1.0474
KEGG_NICOTINATE_AND_NICOTINAMIDE_METABOLISM	1.0093
KEGG_NOD_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.9794
KEGG_NON_HOMOLOGOUS_END_JOINING	0.9661
KEGG_NON_SMALL_CELL_LUNG_CANCER	1.1076
KEGG_NOTCH_SIGNALING_PATHWAY	0.9207
KEGG_NUCLEOTIDE_EXCISION_REPAIR	0.9820
KEGG_OOCYTE_MEIOSIS	0.9581
KEGG_OXIDATIVE_PHOSPHORYLATION	0.9219
KEGG_P53_SIGNALING_PATHWAY	1.0437
KEGG_PANCREATIC_CANCER	0.7924
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION	0.9326
KEGG_PPAR_SIGNALING_PATHWAY	1.0092
KEGG_PRIMARY_IMMUNODEFICIENCY	0.9376
KEGG_PRION_DISEASES	1.0215
KEGG_REGULATION_OF_AUTOPHAGY	0.9630
KEGG_RENAL_CELL_CARCINOMA	1.0650
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.9266
KEGG_RNA_DEGRADATION	0.9573
KEGG_SMALL_CELL_LUNG_CANCER	1.0380
KEGG_SPLICEOSOME	0.9699
KEGG_STEROID_HORMONE_BIOSYNTHESIS	1.0132
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS	1.0154
KEGG_T_CELL_RECEPTOR_SIGNALING_PATHWAY	1.0356
KEGG_TGF_BETA_SIGNALING_PATHWAY	1.0434
KEGG_TIGHT_JUNCTION	1.0918
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.9228
KEGG_TYPE_II_DIABETES_MELLITUS	1.0091
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS	1.0903

KEGG_VIBRIO_CHOLERAE_INFECTION	1.0197
KEGG_VIRAL_MYOCARDITIS	1.0094
	exp(-coef)
age_at_diagnosis	1.0094
smoke_fNever	1.2460
smoke_fUnknown	0.6684
stage_fStage 4	0.3301
prior_med_fPrior medications to MSK	0.6801
prior_med_fUnknown	0.8149
gender	1.3414
KEGG_ABC_TRANSPORTERS	0.9580
KEGG_ACUTE_MYELOID_LEUKEMIA	1.1129
KEGG_ADHERENS_JUNCTION	0.8894
KEGG_ADIPOCYTOKINE_SIGNALING_PATHWAY	0.9256
KEGG_ALLOGRAFT_REJECTION	0.9887
KEGG_AMYOTROPHIC_LATERAL_SCLEROSIS_ALS	0.8867
KEGG_ARRHYTHMOGENIC_RIGHT_VENTRICULAR_CARDIOMYOPATHY_ARVC	1.0285
KEGG_ASTHMA	NA
KEGG_AUTOIMMUNE_THYROID_DISEASE	0.9818
KEGG_AXON_GUIDANCE	1.0672
KEGG_BASAL_TRANSCRIPTION_FACTORS	0.9847
KEGG_BASE_EXCISION_REPAIR	1.0117
KEGG_CHRONIC_MYELOID_LEUKEMIA	0.9120
KEGG_CITRATE_CYCLE_TCA_CYCLE	0.9424
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM	1.0439
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION	1.0309
KEGG_CYTOSOLIC_DNA_SENSING_PATHWAY	0.9576
KEGG_DORSO_VENTRAL_AXIS_FORMATION	0.9812
KEGG_ENDOCYTOSIS	1.0239
KEGG_EPITHELIAL_CELL_SIGNALING_IN_HELICOBACTER_PYLORI_INFECTION	1.1393
KEGG_ERBB_SIGNALING_PATHWAY	1.0384
KEGG_FC_EPSILON_RI_SIGNALING_PATHWAY	0.8843
KEGG_HEDGEHOG_SIGNALING_PATHWAY	1.0441
KEGG_HUNTINGTONS_DISEASE	0.9675
KEGG_HYPERTROPHIC_CARDIOMYOPATHY_HCM	1.0206
KEGG_INSULIN_SIGNALING_PATHWAY	1.2182
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION	1.0128
KEGG_LYSINE_DEGRADATION	1.0212
KEGG_MAPK_SIGNALING_PATHWAY	0.9519
KEGG_MTOR_SIGNALING_PATHWAY	0.8617
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY	0.9547
KEGG_NICOTINATE_AND_NICOTINAMIDE_METABOLISM	0.9908
KEGG_NOD_LIKE_RECEPTOR_SIGNALING_PATHWAY	1.0210

KEGG_NON_HOMOLOGOUS_END_JOINING	1.0351
KEGG_NON_SMALL_CELL_LUNG_CANCER	0.9028
KEGG_NOTCH_SIGNALING_PATHWAY	1.0861
KEGG_NUCLEOTIDE_EXCISION_REPAIR	1.0183
KEGG_OOCYTE_MEIOSIS	1.0437
KEGG_OXIDATIVE_PHOSPHORYLATION	1.0848
KEGG_P53_SIGNALING_PATHWAY	0.9581
KEGG_PANCREATIC_CANCER	1.2620
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION	1.0723
KEGG_PPAR_SIGNALING_PATHWAY	0.9909
KEGG_PRIMARY_IMMUNODEFICIENCY	1.0666
KEGG_PRION_DISEASES	0.9789
KEGG_REGULATION_OF_AUTOPHAGY	1.0385
KEGG_RENAL_CELL_CARCINOMA	0.9389
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHWAY	1.0793
KEGG_RNA_DEGRADATION	1.0446
KEGG_SMALL_CELL_LUNG_CANCER	0.9634
KEGG_SPLICEOSOME	1.0310
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.9870
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS	0.9848
KEGG_T_CELL_RECEPTOR_SIGNALING_PATHWAY	0.9656
KEGG_TGF_BETA_SIGNALING_PATHWAY	0.9584
KEGG_TIGHT_JUNCTION	0.9160
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY	1.0836
KEGG_TYPE_II_DIABETES_MELLITUS	0.9910
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS	0.9172
KEGG_VIBRIO_CHOLERAE_INFECTION	0.9807
KEGG_VIRAL_MYOCARDITIS	0.9906

	lower .95
age_at_diagnosis	0.9877
smoke_fNever	0.7368
smoke_fUnknown	1.3114
stage_fStage 4	2.8315
prior_med_fPrior medications to MSK	1.3502
prior_med_fUnknown	1.1051
gender	0.6987
KEGG_ABC_TRANSPORTERS	1.0075
KEGG_ACUTE_MYELOID_LEUKEMIA	0.8231
KEGG_ADHERENS_JUNCTION	1.0500
KEGG_ADIPOCYTOKINE_SIGNALING_PATHWAY	1.0342
KEGG_ALLOGRAFT_REJECTION	0.9718
KEGG_AMYOTROPHIC_LATERAL_SCLEROSIS_ALS	1.0210
KEGG_ARRHYTHMOGENIC_RIGHT_VENTRICULAR_CARDIOMYOPATHY_ARVC	0.9282

KEGG_ASTHMA	NA
KEGG_AUTOIMMUNE_THYROID_DISEASE	0.9849
KEGG_AXON_GUIDANCE	0.8927
KEGG_BASAL_TRANSCRIPTION_FACTORS	0.9927
KEGG_BASE_EXCISION_REPAIR	0.9486
KEGG_CHRONIC_MYELOID_LEUKEMIA	0.9833
KEGG_CITRATE_CYCLE_TCA_CYCLE	1.0155
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM	0.9261
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION	0.9189
KEGG_CYTOSOLIC_DNA_SENSING_PATHWAY	1.0029
KEGG_DORSO_VENTRAL_AXIS_FORMATION	0.9496
KEGG_ENDOCYTOSIS	0.9251
KEGG_EPITHELIAL_CELL_SIGNALING_IN_HELICOBACTER_PYLORI_INFECTION	0.8250
KEGG_ERBB_SIGNALING_PATHWAY	0.8862
KEGG_FC_EPSILON_RI_SIGNALING_PATHWAY	1.0259
KEGG_HEDGEHOG_SIGNALING_PATHWAY	0.9250
KEGG_HUNTINGTONS_DISEASE	0.9413
KEGG_HYPERTROPHIC_CARDIOMYOPATHY_HCM	0.9471
KEGG_INSULIN_SIGNALING_PATHWAY	0.7440
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION	0.9506
KEGG_LYSINE_DEGRADATION	0.9477
KEGG_MAPK_SIGNALING_PATHWAY	0.9884
KEGG_MTOR_SIGNALING_PATHWAY	1.0909
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY	0.9509
KEGG_NICOTINATE_AND_NICOTINAMIDE_METABOLISM	0.9757
KEGG_NOD_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.9441
KEGG_NON_HOMOLOGOUS_END_JOINING	0.9362
KEGG_NON_SMALL_CELL_LUNG_CANCER	0.9797
KEGG_NOTCH_SIGNALING_PATHWAY	0.8658
KEGG_NUCLEOTIDE_EXCISION_REPAIR	0.9424
KEGG_OOCYTE_MEIOSIS	0.9233
KEGG_OXIDATIVE_PHOSPHORYLATION	0.8766
KEGG_P53_SIGNALING_PATHWAY	0.9896
KEGG_PANCREATIC_CANCER	0.7091
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION	0.8849
KEGG_PPAR_SIGNALING_PATHWAY	0.9765
KEGG_PRIMARY_IMMUNODEFICIENCY	0.9009
KEGG_PRION_DISEASES	0.9811
KEGG_REGULATION_OF_AUTOPHAGY	0.9335
KEGG_RENAL_CELL_CARCINOMA	0.9745
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.8878
KEGG_RNA_DEGRADATION	0.9266
KEGG_SMALL_CELL_LUNG_CANCER	0.9596

KEGG_SPLICEOSOME	0.9385
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.9760
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS	0.9645
KEGG_T_CELL_RECEPTOR_SIGNALING_PATHWAY	0.9573
KEGG_TGF_BETA_SIGNALING_PATHWAY	0.9871
KEGG_TIGHT_JUNCTION	1.0284
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY	0.8520
KEGG_TYPE_II_DIABETES_MELLITUS	0.9450
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS	1.0533
KEGG_VIBRIO_CHOLERAE_INFECTION	0.9817
KEGG_VIRAL_MYOCARDITIS	0.9666

	upper .95
age_at_diagnosis	0.9937
smoke_fNever	0.8741
smoke_fUnknown	1.7066
stage_fStage 4	3.2415
prior_med_fPrior medications to MSK	1.6015
prior_med_fUnknown	1.3627
gender	0.7954
KEGG_ABC_TRANSPORTERS	1.0814
KEGG_ACUTE_MYELOID_LEUKEMIA	0.9809
KEGG_ADHERENS_JUNCTION	1.2040
KEGG_ADIPOCYTOKINE_SIGNALING_PATHWAY	1.1287
KEGG_ALLOGRAFT_REJECTION	1.0528
KEGG_AMYOTROPHIC_LATERAL_SCLEROSIS_ALS	1.2457
KEGG_ARRHYTHMOGENIC_RIGHT_VENTRICULAR_CARDIOMYOPATHY_ARVC	1.0185
KEGG_ASTHMA	NA
KEGG_AUTOIMMUNE_THYROID_DISEASE	1.0533
KEGG_AXON_GUIDANCE	0.9836
KEGG_BASAL_TRANSCRIPTION_FACTORS	1.0388
KEGG_BASE_EXCISION_REPAIR	1.0299
KEGG_CHRONIC_MYELOID_LEUKEMIA	1.2227
KEGG_CITRATE_CYCLE_TCA_CYCLE	1.1089
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM	0.9910
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION	1.0240
KEGG_CYTOSOLIC_DNA_SENSING_PATHWAY	1.0873
KEGG_DORSO_VENTRAL_AXIS_FORMATION	1.0937
KEGG_ENDOCYTOSIS	1.0311
KEGG_EPITHELIAL_CELL_SIGNALING_IN_HELICOBACTER_PYLORI_INFECTION	0.9339
KEGG_ERBB_SIGNALING_PATHWAY	1.0465
KEGG_FC_EPSILON_RI_SIGNALING_PATHWAY	1.2465
KEGG_HEDGEHOG_SIGNALING_PATHWAY	0.9917
KEGG_HUNTINGTONS_DISEASE	1.1349

```
KEGG_HYPERTROPHIC_CARDIOMYOPATHY_HCM                        1.0136
KEGG_INSULIN_SIGNALING_PATHWAY                              0.9057
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION           1.0255
KEGG_LYSINE_DEGRADATION                                     1.0118
KEGG_MAPK_SIGNALING_PATHWAY                                 1.1166
KEGG_MTOR_SIGNALING_PATHWAY                                 1.2347
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY              1.1537
KEGG_NICOTINATE_AND_NICOTINAMIDE_METABOLISM                 1.0440
KEGG_NOD_LIKE_RECEPTOR_SIGNALING_PATHWAY                    1.0161
KEGG_NON_HOMOLOGOUS_END_JOINING                             0.9969
KEGG_NON_SMALL_CELL_LUNG_CANCER                             1.2522
KEGG_NOTCH_SIGNALING_PATHWAY                                0.9791
KEGG_NUCLEOTIDE_EXCISION_REPAIR                             1.0233
KEGG_OOCYTE_MEIOSIS                                         0.9942
KEGG_OXIDATIVE_PHOSPHORYLATION                              0.9695
KEGG_P53_SIGNALING_PATHWAY                                  1.1009
KEGG_PANCREATIC_CANCER                                      0.8855
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION                  0.9828
KEGG_PPAR_SIGNALING_PATHWAY                                 1.0429
KEGG_PRIMARY_IMMUNODEFICIENCY                               0.9757
KEGG_PRION_DISEASES                                         1.0635
KEGG_REGULATION_OF_AUTOPHAGY                                0.9934
KEGG_RENAL_CELL_CARCINOMA                                   1.1640
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHWAY                  0.9670
KEGG_RNA_DEGRADATION                                        0.9891
KEGG_SMALL_CELL_LUNG_CANCER                                 1.1229
KEGG_SPLICEOSOME                                            1.0024
KEGG_STEROID_HORMONE_BIOSYNTHESIS                           1.0518
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS                           1.0690
KEGG_T_CELL_RECEPTOR_SIGNALING_PATHWAY                      1.1204
KEGG_TGF_BETA_SIGNALING_PATHWAY                             1.1029
KEGG_TIGHT_JUNCTION                                         1.1590
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY                   0.9995
KEGG_TYPE_II_DIABETES_MELLITUS                              1.0775
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS                         1.1286
KEGG_VIBRIO_CHOLERAE_INFECTION                              1.0591
KEGG_VIRAL_MYOCARDITIS                                      1.0542

Concordance= 0.715  (se = 0.004 )
Likelihood ratio test= 1909  on 67 df,   p=<2e-16
Wald test            = 1893  on 67 df,   p=<2e-16
Score (logrank) test = 2026  on 67 df,   p=<2e-16
```

C-index

```
# Setup CV
set.seed(629)
K <- 5
n_path <- nrow(final_model_data_path)
fold_id_path <- sample(rep(1:K, length.out = n_path))
cv_cindex_post_lasso_path <- numeric(K)

for(k in 1:K){

  train_data_path <- final_model_data_path[fold_id_path != k, ]
  test_data_path  <- final_model_data_path[fold_id_path == k, ]

  # Fit the post-LASSO model on the training fold
  # (Using iter.max = 100 to prevent convergence warnings on smaller folds)
  fit_path <- coxph(
    Surv(entry_time, exit_time, event) ~ .,
    data = train_data_path,
    control = coxph.control(iter.max = 100)
  )

  # Predict risk scores on the test fold
  test_data_path$risk_score <- predict(fit_path, newdata = test_data_path, type = "lp", na.ac

  # Compute C-index on the test fold
  c_obj_path <- concordance(
    Surv(entry_time, exit_time, event) ~ risk_score,
    data = test_data_path,
    reverse = TRUE
  )

  cv_cindex_post_lasso_path[k] <- c_obj_path$concordance
}

# Print results
cat("Pathway Model Mean CV C-index:", mean(cv_cindex_post_lasso_path, na.rm = TRUE), "\n")
```

```
Pathway Model Mean CV C-index: 0.707944
```

```r
cat("Pathway Model SD CV C-index:", sd(cv_cindex_post_lasso_path, na.rm = TRUE), "\n")
```

```
Pathway Model SD CV C-index: 0.00664873
```

Time-Dependent ROC Curve

```r
library(timeROC)

# 1. Generate the risk scores (linear predictors) from your final pathway model
final_model_data_path$risk_score <- predict(clean_cox_model_path, type = "lp")

# 2. Define the time horizon (80th percentile of follow-up time)
time_horizon_path <- quantile(final_model_data_path$exit_time, 0.80)

# 3. Calculate the Time-dependent ROC
roc_pathway <- timeROC(
  T = final_model_data_path$exit_time,
  delta = final_model_data_path$event,
  marker = final_model_data_path$risk_score,
  cause = 1,
  times = time_horizon_path,
  iid = TRUE # Keeps the data needed to calculate Confidence Intervals
)

# 4. Extract the Area Under the Curve (AUC) and Confidence Intervals
auc_value_path <- roc_pathway$AUC[2] # Index 2 corresponds to time_horizon_path
se_value_path <- roc_pathway$inference$vect_sd_1[2]

# Calculate 95% CI
ci_lower_path <- auc_value_path - 1.96 * se_value_path
ci_upper_path <- auc_value_path + 1.96 * se_value_path

cat("Time Horizon:", round(time_horizon_path, 2), "\n")
```

```
Time Horizon: 66.47
```

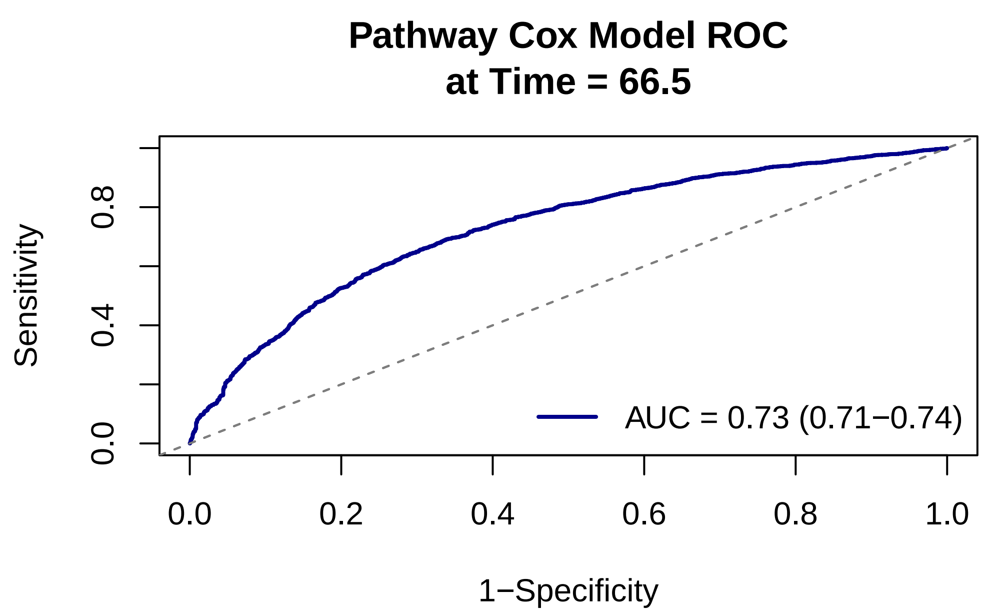
```r
cat("Pathway AUC at Time Horizon:", round(auc_value_path, 3), "\n")
```

```
Pathway AUC at Time Horizon: 0.729
```

```
cat("95% CI: [", round(ci_lower_path, 3), "-", round(ci_upper_path, 3), "]\n\n")
```

```
95% CI: [ 0.714 - 0.744 ]
```

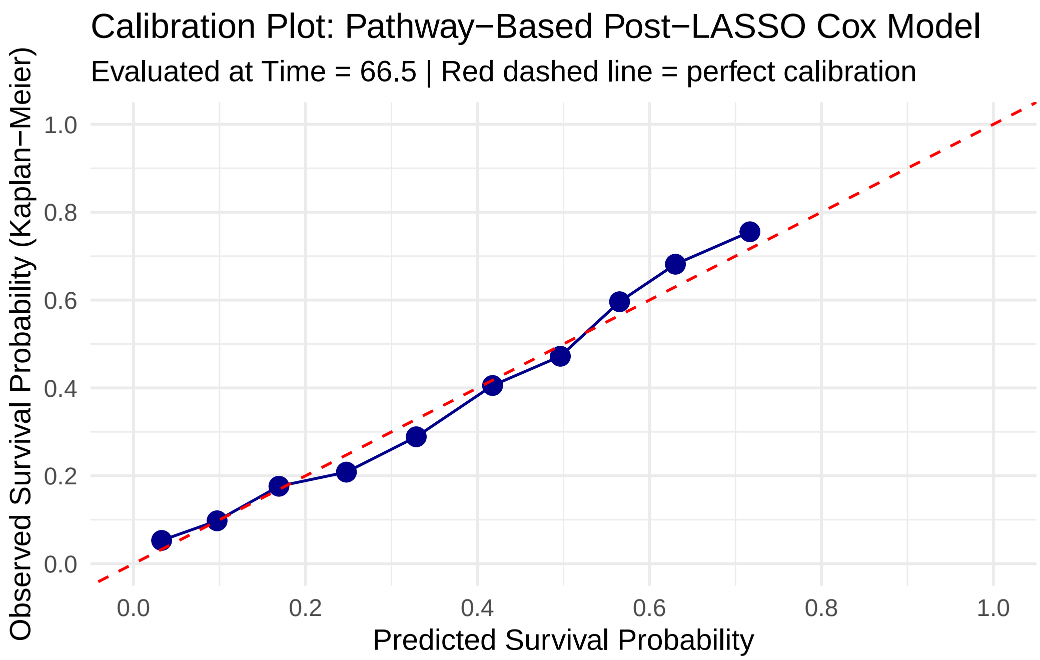
```
# 5. Plot the ROC Curve
{plot(roc_pathway,
     time = time_horizon_path,
     col = "darkblue", # Changed color to visually distinguish from your gene model
     lwd = 2,
     title = FALSE)

# Add a reference line and a title
abline(0, 1, lty = 2, col = "gray50")
title(main = paste("Pathway Cox Model ROC\nat Time =", round(time_horizon_path, 1)))

# Add a legend with the AUC value
legend("bottomright",
       legend = paste0("AUC = ", round(auc_value_path, 2),
                       " (", round(ci_lower_path, 2), "-", round(ci_upper_path, 2), ")"),
       col = "darkblue",
       lwd = 2,
       bty = "n")}
```



calibration plot

```
library(ggplot2)
library(dplyr)
library(survival)

# 1. Define the time horizon (80th percentile of follow-up time, same as ROC)
time_horizon_path <- quantile(final_model_data_path$exit_time, 0.80)

# 2. Calculate predicted survival probabilities for the entire cohort at the time horizon
sf_final_path <- survfit(clean_cox_model_path, newdata = final_model_data_path)
pred_surv_path <- summary(sf_final_path, times = time_horizon_path)$surv
pred_surv_path <- as.numeric(pred_surv_path)

# 3. Add predictions to the dataset
final_model_data_path$pred_surv <- pred_surv_path

# 4. Create 10 risk groups (deciles) based on predicted survival
# Note: If your dataset is small, you might want to use 5 groups (ntile(..., 5)) instead
final_model_data_path$decile <- ntile(final_model_data_path$pred_surv, 10)

# 5. Calculate observed Kaplan-Meier estimates for each decile
calibration_data_path <- final_model_data_path %>%
  group_by(decile) %>%
  group_modify(~ {

    km_fit <- survfit(Surv(entry_time, exit_time, event) ~ 1, data = .x)

    # extend = TRUE ensures we get an estimate even if the last event in a decile
    # happened slightly before the time horizon
    km_summary <- summary(km_fit,
                          times = time_horizon_path,
                          extend = TRUE)

    data.frame(
      mean_pred = mean(.x$pred_surv, na.rm = TRUE),
      km_est = km_summary$surv
    )
  })

# 6. Plot the calibration curve
ggplot(calibration_data_path, aes(x = mean_pred, y = km_est)) +
```

```
  geom_point(size = 3, color = "darkblue") +
  geom_line(color = "darkblue") +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "red") +
  scale_x_continuous(limits = c(0, 1), breaks = seq(0, 1, 0.2)) +
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, 0.2)) +
  labs(
    x = "Predicted Survival Probability",
    y = "Observed Survival Probability (Kaplan-Meier)",
    title = "Calibration Plot: Pathway-Based Post-LASSO Cox Model",
    subtitle = paste("Evaluated at Time =", round(time_horizon_path, 1), "| Red dashed line =
  ) +
  theme_minimal()
```



2. Left-truncated random forest

Data Cleaning and Train/Test Split (80/20)

```
# Clean the dataset to ensure the forest only sees true predictors
rsf_data <- final_model_data_path %>%
  select(-any_of(c("risk_score", "pred_surv", "decile"))) %>%
```

```r
  as.data.frame()

set.seed(629)
N_total <- nrow(rsf_data)
train_idx <- sample(1:N_total, size = floor(0.8 * N_total))

train_data_rsf <- rsf_data[train_idx, ]
test_data_rsf  <- rsf_data[-train_idx, ]

cat("Train patients:", nrow(train_data_rsf), "\n")
```

```
Train patients: 6241
```

```r
cat("Test patients:", nrow(test_data_rsf), "\n")
```

```
Test patients: 1561
```

Fit the LTRC Forest on TRAIN Data

```r
predictor_names <- setdiff(colnames(rsf_data), c("entry_time", "exit_time", "event"))

# Build the formula dynamically
rsf_formula <- as.formula(
  paste("Surv(entry_time, exit_time, event) ~", paste(predictor_names, collapse = " + "))
)

# Fit the forest using ONLY the training data
set.seed(629)
final_rsf_model <- ltrcrrf(
  formula = rsf_formula,
  data = train_data_rsf,
  ntree = 500,
  nodesize = 20,
  mtry = 3,
  nsplit = 10
)
```

Evaluation on Independent TEST Data

c-index

```
# Define the time horizon based on the training set follow-up
time_horizon_path <- quantile(train_data_rsf$exit_time, 0.80)

# Predict survival probabilities on the TEST set
test_pred_obj <- predictProb(
  final_rsf_model,
  newdata = test_data_rsf,
  time.eval = time_horizon_path
)

# Add predictions and risk scores to the test dataset
test_data_rsf$pred_surv <- as.numeric(test_pred_obj$survival.probs)
test_data_rsf$rsf_risk <- 1 - test_data_rsf$pred_surv

c_obj_rsf <- concordance(
  Surv(entry_time, exit_time, event) ~ rsf_risk,
  data = test_data_rsf,
  reverse = TRUE
)

cat("\nOut-of-Sample C-index:", round(c_obj_rsf$concordance, 3), "\n")
```

```
Out-of-Sample C-index: 0.674
```

Time-Dependent ROC on TEST

```
roc_rsf <- timeROC(
  T = test_data_rsf$exit_time,
  delta = test_data_rsf$event,
  marker = test_data_rsf$rsf_risk,
  cause = 1,
  times = time_horizon_path,
  iid = TRUE
)

auc_value_rsf <- roc_rsf$AUC[2]
```

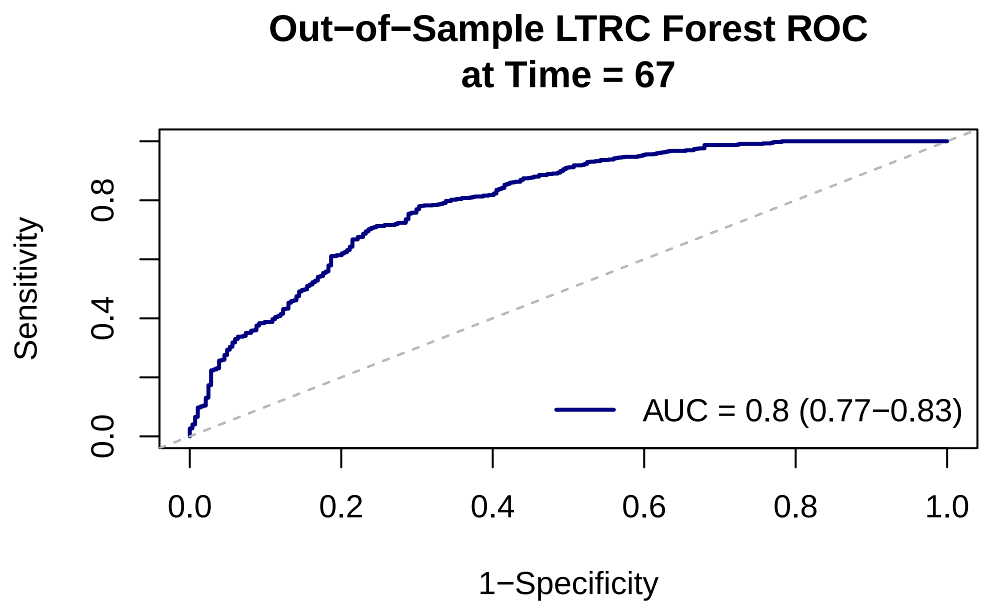
```r
se_value_rsf <- roc_rsf$inference$vect_sd_1[2]
ci_lower_rsf <- auc_value_rsf - 1.96 * se_value_rsf
ci_upper_rsf <- auc_value_rsf + 1.96 * se_value_rsf

cat("Out-of-Sample AUC at Time Horizon (", round(time_horizon_path, 1), "):", round(auc_value
```

```
Out-of-Sample AUC at Time Horizon ( 67 ): 0.802
```

```r
cat("95% CI: [", round(ci_lower_rsf, 3), "-", round(ci_upper_rsf, 3), "]\n\n")
```

```
95% CI: [ 0.77 - 0.833 ]
```

```r
plot(roc_rsf,
     time = time_horizon_path,
     col = "navy",
     lwd = 2,
     title = FALSE)

abline(0, 1, lty = 2, col = "gray")
title(main = paste("Out-of-Sample LTRC Forest ROC\nat Time =", round(time_horizon_path, 1)))

legend("bottomright",
       legend = paste0("AUC = ", round(auc_value_rsf, 2),
                       " (", round(ci_lower_rsf, 2), "-", round(ci_upper_rsf, 2), ")"),
       col = "navy",
       lwd = 2,
       bty = "n")
```



Calibration Plot on TEST

```
# Using 5 groups (quintiles) instead of 10 deciles because the test set is 20% of the total s
test_data_rsf$decile <- ntile(test_data_rsf$pred_surv, 5)

calibration_data_rsf <- test_data_rsf %>%
  group_by(decile) %>%
  group_modify(~ {

    km_fit <- survfit(Surv(entry_time, exit_time, event) ~ 1, data = .x)
    km_summary <- summary(km_fit,
                          times = time_horizon_path,
                          extend = TRUE)

    data.frame(
      mean_pred = mean(.x$pred_surv, na.rm = TRUE),
      km_est = km_summary$surv
    )
  })

ggplot(calibration_data_rsf, aes(x = mean_pred, y = km_est)) +
  geom_point(size = 3, color = "navy") +
```

```
geom_line(color = "navy") +
geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "red") +
scale_x_continuous(limits = c(0, 1), breaks = seq(0, 1, 0.2)) +
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, 0.2)) +
labs(
  x = "Predicted Survival Probability",
  y = "Observed Survival Probability (Kaplan-Meier)",
  title = "Out-of-Sample Calibration: LTRC Forest"
) +
theme_minimal()
```